Risen Electric Configurations — Calculation, Performance and Operating Cost

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Summary

At 300 Wh/kg, electrification trades ICE range for lower variable cost:

- **Range and endurance.** The electric variants fly 215–384 km and 1.5–2.8 h with a 30-min reserve, against about 2,800 km for the ICE Risen.
- **Cost.** They reach cost parity with ICE at 1,200 h/yr, but cost 10–11 % more at private-owner utilisation (150 h/yr).

All figures are produced by `risen_eas.py`, assume a 1,500 m ISA cruise with a 30-min reserve at V_md, and are beginning-of-life unless noted.

Metric	Risen 912iS (ICE)	E600 2-seat electric	E750 2-seat electric	Option A, 2 occ. (both packs)	Option A, 4 occ. (main pack)
MTOM / take-off mass [kg]	600 / 600	600 / 600	750 / 750	1,177.9 / 1,137.9	1,177.9 / 1,177.9
Energy carried	100.7 L fuel	140 kg / 42.0 kWh	224 kg / 67.2 kWh	414 kg / 124.2 kWh	274 kg / 82.2 kWh
Range at about 216–220 km/h [km]	2,815 (datasheet, FL90)	215	302	384	215
Range at end of pack life (80 %) [km]	—	153	223	289	153
Best range at V_md [km]	—	233	329	426	233
Endurance at V_mp [h]	≈ 12.8	1.50	2.14	2.82	1.50
Cost at 150 h/yr [₹/h]	36,516	40,250	49,367	76,055 (50/50 mix)	76,055 (50/50 mix)
Cost at 1,200 h/yr [₹/h]	8,883	8,723	10,523	15,866 (50/50 mix)	15,866 (50/50 mix)
Cost at 1,200 h/yr [₹/km]	40.4	41.8	50.5	76.6 (mix)	76.6 (mix)
Cost at 1,200 h/yr [₹/seat-km]	20.2	20.9	25.2	37.8 (2-occ. only)	19.8 (4-occ. only)

Configurations and mass breakdown

All three electric variants keep the Risen's aerodynamic configuration and wing loading (61.9 kg/m²). The Rotax 912iS is replaced by a 2.5 kW/kg electric powertrain and a 300 Wh/kg pack.

- **Risen 912iS (ICE)** is the datasheet baseline: 600 kg MTOM, 325 kg empty, 73.5 kW Rotax 912iS, 100.7 L fuel.
- **E600** is the same airframe and MTOM with electric power, two occupants and the battery closing the mass (fixed-MTOM mode).
- **E750** scales the airframe to 750 kg MTOM at constant wing loading, with two occupants (fixed-MTOM mode).
- **Option A** is a four-seat airframe sized for a 274 kg main pack plus a 140 kg removable pack. The removable pack is fitted only when two occupants fly. MTOM is solved from the binding case: four occupants with the main pack only.

Item	Risen 912iS	E600	E750	Option A
Wing area / span	9.70 m² / 8.99 m	9.70 m² / 8.99 m	12.12 m² / 10.05 m	19.04 m² / 12.60 m
Installed shaft power [kW]	73.5	75.0	93.8	147.2
Structure [kg]	190.0	190.0	242.9	399.9
Fixed equipment [kg]	45.0	45.0	48.0	61.0
Propeller [kg]	15.0	15.0	17.5	24.1
Powerplant [kg]	≈ 75 (Rotax installed)	30.0	37.5	58.9
Operating empty mass excl. battery [kg]	**325 (datasheet)**	**280.0**	**346.0**	**543.9**
Payload [kg]	180 (2 × 85 + 10)	180	180	180 or 360 (4 × 85 + 20)
Battery or fuel [kg]	72.5 fuel	140.0	224.0	274 + 140
Battery mass fraction	—	23.3 %	29.9 %	36.4 % (2 occ.) / 23.3 % (4 occ.)
Capex: airframe + powertrain, landed [₹ Cr]	3.29	3.56	4.24	6.36
Capex: packs [₹ L]	—	29.4	47.0	57.5 + 29.4

Calculation approach applied

The method follows the companion document *Electric Aircraft Weight–Performance–Cost Estimation Method*. Its core equations are below, followed by every configuration's intermediate values.

$$P_{bat}=\frac{DV}{\eta},\qquad R_{ideal}=\frac{E_{use}\,\eta}{W}\left(\frac{L}{D}\right)_{max},\qquad R=d_{cl}+\frac{E_{use}-E_{ovh}-E_{res}}{P_{bat}(V)}V+d_{ds}$$

Polar. The polar is calibrated on the ICE datasheet's maximum-speed point (73.5 kW at 175 kt), which gives an ICE C_{D0} of 0.0128. This polar predicts 56.5 kW at the datasheet's 75 %-power point (159 kt), against 55.1 kW, confirming it. The electric C_{D0} is 0.0120, AR 8.33, e 0.80, giving (L/D)max = 20.9. The chain efficiency η = 0.708.

Worked example (E600, 216 km/h).

Step	Value
Usable energy, E_use = 42.0 × 0.90	37.8 kWh
Sortie overhead, E_ovh	6.82 kWh
Reserve, E_res = 19.0 kW at V_md × 0.5 h	9.52 kWh
Cruise energy, 37.8 − 6.82 − 9.52	21.5 kWh
Cruise time at 26.4 kW	0.81 h
Cruise distance (0.81 h × 216 km/h)	175.8 km
Plus climb 9.5 km and descent 29.5 km	**R = 214.7 km**

Intermediate value (1,500 m ISA)	E600	E750	Option A, 2 occ.	Option A, 4 occ.
Weight W [N]	5,884	7,355	11,159	11,551
V_md / V_mp [km/h]	172 / 131	172 / 131	169 / 129	172 / 131
P_bat at 216 km/h [kW]	26.4	33.0	50.8	51.8
Energy per km at 216 km/h [Wh/km]	122	153	235	240
Usable energy, beginning / end of life [kWh]	37.8 / 30.2	60.5 / 48.4	111.8 / 89.4	74.0 / 59.2
Sortie overhead E_ovh [kWh]	6.82	8.46	12.66	13.11
Reserve E_res [kWh]	9.52	11.90	17.74	18.68
Rate of climb [m/s]	6.29	6.29	6.60	6.29
Ideal range R_ideal [km]	342	438	533	341
Ideal endurance T_ideal [h]	2.26	2.90	3.59	2.26

Option A four-up returns the same ranges as E600. Both have a 23.3 % battery mass fraction and equal wing loading. Range depends only on energy per unit weight, (L/D) and η, so absolute size cancels out.

Performance metrics

Option A with both packs is the longest-range electric case: 384 km at 216 km/h and 289 km at end of pack life, both with a 30-min reserve. Option A four-up has only 2.0C of take-off margin, the same as E600, which makes it the binding power case for cell selection.

Metric	E600	E750	Option A, 2 occ.	Option A, 4 occ.
Range at V_md (≈172 km/h), 30-min reserve [km]	233	329	426	233
Range at 216 km/h, 30-min reserve [km]	215	302	384	215
Range at 216 km/h, no reserve [km]	293	380	460	293
Range at 216 km/h, end of life (80 %), 30-min reserve [km]	153	223	289	153
Range at 252 km/h, 30-min reserve [km]	188	262	329	188
Endurance at V_mp, 30-min reserve [h]	1.50	2.14	2.82	1.50
Endurance at V_mp, no reserve [h]	2.07	2.71	3.39	2.07
Endurance at V_mp, end of life, 30-min reserve [h]	1.05	1.56	2.11	1.05
Take-off / climb C-rate	2.02 / 1.75	1.58 / 1.37	1.34 / 1.16	2.02 / 1.75
Pack specific power at take-off [W/kg]	605	473	402	607

The ICE Risen ranges below come from the datasheet, converted to SI. They include a 30-min reserve at long-range power.

ICE power setting	Sea level: TAS / range	FL90 (2,743 m): TAS / range
Long range (7.6 L/h at SL)	193 km/h / 2,334 km	220 km/h / 2,815 km
75 % power	295 km/h / 1,519 km	300 km/h / 1,926 km
Maximum power	324 km/h / 1,120 km	324 km/h / 1,426 km

Operating cost metrics

Utilisation dominates cost. Each configuration's cost per hour falls 76–79 % from 150 to 1,200 h/yr. Against its ICE equivalent, electric is 10–11 % dearer at 150 h/yr and 2–6 % cheaper at 1,200 h/yr.

Sortie assumptions: 1.0 h airborne at 216 km/h with two occupants (209 km); 0.7 h with four occupants in Option A (144 km). The mix is by flight hours. Figures are full-recovery cost with no margin.

Cost per flight hour [₹/h]	150 h/yr	600 h/yr	1,200 h/yr	₹/km at 1,200 h/yr	₹/seat-km at 1,200 h/yr
Risen 912iS (ICE, mogas)	36,516	12,282	8,883	40.4	20.2
E600	40,250	12,812	8,723	41.8	20.9
E750	49,367	15,464	10,523	50.5	25.2
Option A, 100 % two occupants	76,042	23,343	15,807	75.7	37.8
Option A, 50/50 mix	76,055	23,552	15,866	76.6	25.6
Option A, 100 % four occupants	76,067	24,075	16,233	79.0	19.8
1,178 kg ICE comparator, mogas (50/50 mix)	68,773	23,253	16,894	76.8	25.6
1,178 kg ICE comparator, avgas ₹200/L	—	—	18,240	82.9	27.6

Cost build-up at 1,200 h/yr [₹ per flight hour]

Element	Risen 912iS (ICE)	E600	E750	Option A (50/50)	1,178 kg ICE
Depreciation	—	2,075	2,474	3,712	—
Cost of capital	—	1,845	2,244	3,428	—
Insurance	—	802	982	1,507	—
Hangar + annual inspection	—	467	537	736	—
Charger / pack-handling infrastructure	—	283	354	627	—
Fixed subtotal	**4,680**	**5,472**	**6,591**	**10,011**	**8,815**
Energy (fuel or electricity)	1,028	374	467	730	2,019
Battery wear	—	1,427	1,780	2,784	—
Engine overhaul reserve	1,375	—	—	—	3,300
Maintenance (motor, propeller, airframe)	1,800	1,450	1,686	2,342	2,760

Element	Risen 912iS (ICE)	E600	E750	Option A (50/50)	1,178 kg ICE
Total	**8,883**	**8,723**	**10,523**	**15,866**	**16,894**
Pack life at 1,200 h/yr	—	1.7 yr	2.2 yr	main 2.1 yr / removable 5.3 yr	—

At 150 h/yr every pack is calendar-limited (8 yr), so battery wear becomes a fixed cost. That is why electric is dearer at low utilisation. Across 1,000–2,500 cycles, the Option A 1,200 h/yr cost spans ₹17,260–14,780/h.

Verification and corrections

The code, this document and the method document agree on all 61 checked metrics, after two code iterations and two corrections to figures quoted earlier in the conversation.

`python3 risen_eas.py --verify` compares each published metric to the code output. Tolerances are 0.2–0.5 % for masses, 1–1.5 % for ranges, endurances and costs, and 2–5 % for calibration points. The largest residual deviation is 0.23 % (E750 take-off C-rate).

Iteration	Result	Finding	Fix
1	37 / 61 pass	Solved-MTOM closure summed both packs with the four-occupant payload, giving 1,430 kg instead of 1,178 kg. All Option A metrics failed.	Approach clarified and code fixed: MTOM is set by the heaviest loading case, max(W_p + packs fitted).
1	Polar check failed	C_D0 = 0.012 had been described as datasheet-calibrated. It reproduces 68.9 kW at 175 kt, not 73.5 kW.	Approach clarified: calibrated ICE C_D0 = 0.0128 (75 %-power cross-check +2.5 %). The electric 0.012 is now stated as ICE C_D0 less 6.6 % cooling/exhaust drag. No performance figure changes.
2	61 / 61 pass	—	—

Corrections to earlier-quoted figures.

Metric	Earlier	Converged	Reason
E600 cost at 150 h/yr	₹40,300/h	₹40,250/h	Energy per hour now from the sortie model (27.5 kWh), not cruise power plus full overhead (33.2 kWh)
E600 cost per km at 1,200 h/yr	₹41.9/km	₹41.8/km	Rounding of sortie distance
E750 cost per km at 1,200 h/yr	₹50.4/km	₹50.5/km	Rounding (code value 50.46)
Wing loading (750 kg answer)	618.6 kg/m²	61.9 kg/m²	Typographical error; calculations were unaffected